Sharp bounds on population-order reversal in autonomous qubit reset

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Population measurements reveal only part of quantum relaxation, because coherence can preserve information that is absent from the populations. Near qubit reset, this raises the question of how strongly autonomous dynamics can reverse population order and what stationary coherence such a reversal requires. For reset toward $|1\rangle$, let $a = P(0|0)$ and $\epsilon = P(0|1)$. Over all time-independent qubit Lindblad generators and their probability closure, the largest reversal at fixed actual disturbance obeys $G(\epsilon) = \max(\epsilon - a) \sim (4/e)\epsilon/\log(1/\epsilon)$. Requiring a stationary state diagonal in the preparation–measurement basis reduces the sharp scale to $G_{\mathrm{diag}}(\epsilon) \sim (64/e^2)\epsilon/\log^2(1/\epsilon)$. Uniform upper bounds and explicit single-jump constructions establish both laws. The refined bound quantifies how a measured reversal constrains the coherence and purity of every stationary state. It also implies a physical tradeoff: leading-order optimal population reversal requires a worst-case trace-distance reset error of order $\sqrt{\epsilon}$, whereas incoherent relaxation can approach error ϵ at the same disturbance. Population crossings and a dephasing intervention identify the coherence that carries the remaining input distinguishability. These results connect sharp homogeneous-embedding bounds to the stationary resources and full-state cost of population reversal.

I. INTRODUCTION

Autonomous relaxation uses a fixed open-system generator to approach a stationary state. For a classical two-state system, the order of two initial populations is preserved throughout the relaxation. A qubit has an additional route: population differences can be transferred into coherence and subsequently return to the measured populations. Two initially ordered populations can then cross, even though the quantum evolution remains Markovian. Near reset, both inputs have nearly the target population, so the question concerns the small residual dependence on the input: how large can its reversal be under one fixed generator?

The corresponding homogeneous quantum-embedding problem asks which conditional probability tables can arise from time-independent quantum Markov dynamics. Shahbeigi *et al.* obtained analytic exclusion regions and numerical boundary candidates [1]. Their four-parameter single-jump family provides the antecedent for the attaining dynamics below, but its optimality over all generators remained unresolved. The sharp reset-endpoint bounds derived here apply to the full generator class and to the class admitting a stationary state diagonal in the preparation–measurement basis.

The qualitative role of coherence has established precedents. Quantum fixed-point reachability can exceed classical population reachability [2]; generating coherence and converting it into populations underlies nonclassical measurement statistics [3]; and classical reductions of quantum dynamics can exhibit information backflow through coherence [4]. The present endpoint laws quantify the restriction imposed by a single fixed generator. Direct coherence readout in a tilted stationary eigenbasis and feedback into population yield different optimal logarithmic scales.

The bounds also answer an inverse physical question: what stationary structure is required by an observed population reversal? A nonzero fraction of the unrestricted leading optimum imposes a quantitative lower bound on the coherence of every stationary state and an upper bound on its mixedness. The same coherence gives a lower bound on the full-state reset error. This connects the population boundary to a physical tradeoff between conditional population control and closeness to the target quantum state. Explicit time trajectories explain how the required coherence preserves input distinguishability at a population crossing.

II. MODEL

Prepare a qubit in either $|0\rangle$ or $|1\rangle$, evolve both inputs with the same generator for the same duration, and measure in the preparation basis. Near reset toward $|1\rangle$, write the column-stochastic probability table as

$$T(a, 1-\epsilon) = \begin{pmatrix} a & \epsilon \\ 1-a & 1-\epsilon \end{pmatrix}. \tag{1}$$

Here a is the unwanted output population for input $|0\rangle$, and ϵ is the disturbance of the target input $|1\rangle$. Their difference equals the determinant of the probability table:

$$q := a - \epsilon = \det T. \tag{2}$$

For a diagonal input with unwanted population r, linearity gives

$$\rho_r = r|0\rangle\langle 0| + (1-r)|1\rangle\langle 1|, \qquad P_0(r) = \epsilon + (a-\epsilon)r. \tag{3}$$

Thus $q < 0$ means that an input with more unwanted population ends with less of it. The optimization problem

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is to determine the largest such population-order reversal generated by one fixed qubit Lindbladian.

Fixing the actual measured disturbance ϵ selects a slice of the homogeneous-embedding region. For a realized reversal $\Delta = \epsilon - a > 0$, the worst population error among diagonal inputs remains ϵ, while the error averaged equally over the two basis inputs is $\epsilon - \Delta/2$. This is therefore an optimization of input-conditioned population data. With only an upper allowance $\epsilon \le \delta$ and unrestricted time and rates, ordinary damping toward $|1\rangle$ could already approach perfect reset with $\epsilon = 0$. The two measured probabilities do not specify the response to coherent inputs.

The model includes all time-independent Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) generators [5, 6]

$$\begin{aligned} \mathcal{L}(\rho) &= -i[H,\rho] + \sum_{\mu} \mathcal{D}[J_\mu](\rho), \\ \mathcal{D}[J](\rho) &= J\rho J^\dagger - \tfrac{1}{2}\{J^\dagger J, \rho\}, \end{aligned} \tag{4}$$

on $\mathbb{C}^2$, with a Hermitian Hamiltonian and a finite collection of finite jump matrices. No auxiliary levels are allowed, and the preparation–measurement basis remains fixed. Rates, duration, and Hamiltonian strength are unrestricted.

Identify each table $T(a, 1-\epsilon)$ with the pair $(a, 1-\epsilon)$. Let $\mathcal{Q}_2^{\mathrm{fin}}$ be the tables obtained from $e^{t\mathcal{L}}$ at finite $t \ge 0$, and set

$$\mathcal{Q}_2 = \overline{\mathcal{Q}_2^{\mathrm{fin}}} \subset [0,1]^2. \tag{5}$$

This probability closure implements the arbitrary-accuracy convention of Ref. [1]: each approximant has finite parameters, with no common bound on rates or duration. Rescaling $H \mapsto tH$, $J_\mu \mapsto \sqrt{t} J_\mu$ fixes the observation time at one and retains singular parameter sequences.

At fixed actual disturbance, define

$$G(\epsilon) = \max\{\epsilon - a : (a, 1-\epsilon) \in \mathcal{Q}_2\}. \tag{6}$$

The maximum exists because $\mathcal{Q}_2$ is compact and its fixed-ϵ slice is nonempty: classical relaxation to population ϵ gives $a = \epsilon$ in the closure. In particular, $0 \le G(\epsilon) \le \epsilon$.

Let $\mathcal{Q}_{\mathrm{diag}}$ be the probability closure of finite realizations whose generator admits at least one stationary state diagonal in the preparation–measurement basis. This class permits transient coherence and population–coherence coupling; a unique stationary state and preservation of all diagonal states are not required. Define G_{diag} by (6) with $\mathcal{Q}_2$ replaced by $\mathcal{Q}_{\mathrm{diag}}$.

III. POPULATION-REVERSAL BOUNDS

The two classes have different asymptotic boundaries near perfect reset.

Theorem 1 (Sharp autonomous reset law). For the full time-independent qubit GKSL probability closure,

$$\lim_{\epsilon \downarrow 0} \frac{\log(1/\epsilon)}{\epsilon} G(\epsilon) = \frac{4}{e}. \tag{7}$$

Equivalently, the lower reset boundary is

$$\min\{a : (a, 1-\epsilon) \in \mathcal{Q}_2\} = \epsilon \left[1 - \frac{4/e + o(1)}{\log(1/\epsilon)}\right]. \tag{8}$$

Theorem 2 (Diagonal-stationary-state law). The restricted class has the sharp endpoint asymptotic

$$G_{\mathrm{diag}}(\epsilon) = \left(\frac{64}{e^2} + o(1)\right) \frac{\epsilon}{\log^2(1/\epsilon)}. \tag{9}$$

Both laws imply $a/\epsilon \to 1$ on the optimal boundary. The two input-conditioned errors agree to leading order, while their smaller difference $\epsilon - a$ measures the population-order reversal. This difference is a vanishing fraction of the actual disturbance in the perfect-reset limit.

The two asymptotic laws lie between the classical and time-dependent quantum limits:

$$\begin{aligned} &G_{\mathrm{cl}}(\epsilon) = 0, && G_{\mathrm{diag}}(\epsilon) \sim \frac{64}{e^2} \frac{\epsilon}{\log^2(1/\epsilon)}, \\ &G(\epsilon) \sim \frac{4}{e} \frac{\epsilon}{\log(1/\epsilon)}, && G_{\mathrm{td}}(\epsilon) = \epsilon. \end{aligned} \tag{10}$$

Here G_{cl} and G_{td} denote the same fixed-error optimization, with the same probability-closure convention, for classical two-state dynamics and time-dependent qubit GKSL control, respectively. Even with time-dependent classical rates, $q(t) = \exp[-\int_0^t (k_{01}(s) + k_{10}(s))\, ds] \ge 0$. Time-dependent qubit GKSL control permits every two-state stochastic table [2]. All four comparisons use the same system dimension, preparation–measurement basis, and probability task.

The unrestricted leading optimum also constrains the stationary states of any attaining sequence. For a finite realization with $a \le \epsilon$ sufficiently small, write any stationary state in its eigenbasis as

$$\begin{aligned} &\rho_* = \nu |e\rangle\langle e| + (1-\nu)|g\rangle\langle g|, \\ &0 \le \nu < \tfrac{1}{2}, \qquad p = |\langle 0|g\rangle|^2. \end{aligned} \tag{11}$$

Here ν is the smaller stationary eigenvalue, and p measures the tilt of the dominant eigenvector away from the target state $|1\rangle$.

Corollary 3 (Stationary-state saturation). Consider a sequence of finite qubit GKSL realizations with actual errors a_n, ϵ_n, where $\epsilon_n \downarrow 0$ and

$$\frac{\log(1/\epsilon_n)}{\epsilon_n}(\epsilon_n - a_n) \longrightarrow \frac{4}{e}. \tag{12}$$

For each sufficiently large n, choose any stationary state and let ν_n, p_n be as in (11). Then

$$\frac{\nu_n}{\epsilon_n} \longrightarrow 0, \qquad \frac{p_n}{\epsilon_n} \longrightarrow 1. \tag{13}$$

In the fixed measurement basis this gives

$$|\langle 0|\rho_{*,n}|1\rangle|^2 = (1-2\nu_n)^2 p_n(1-p_n) \sim \epsilon_n,$$
$$1-\mathrm{tr}(\rho_{*,n}^2) = 2\nu_n(1-\nu_n) = o(\epsilon_n). \tag{14}$$

Although stationary coherence vanishes in the reset limit, its square exhausts the leading population-error budget. The purity defect is of smaller order. Section V proves these necessary conditions for sequences of finite realizations.

The relative separation of the two optimal gains is

$$\frac{G_{\mathrm{diag}}(\epsilon)}{G(\epsilon)} \sim \frac{16}{e\log(1/\epsilon)} \longrightarrow 0. \tag{15}$$

Consequently any finite-generator sequence for which $\liminf \log(1/\epsilon)(\epsilon-a)/\epsilon > 0$ eventually admits no stationary state diagonal in the measurement basis: a contrary subsequence would satisfy Theorem 2 and have vanishing scaled gain. Stationary coherence is therefore necessary for a nonvanishing fraction of the unrestricted leading scale, while transient coherence can realize the smaller scale.

The same bound provides quantitative information away from complete saturation. For a finite realization with $\Delta = \epsilon - a \geq 0$, define its normalized reversal and the coherence and impurity of any stationary state by

$$r_\epsilon = \frac{e}{4}\frac{\log(1/\epsilon)}{\epsilon}\Delta, \qquad C_* = |\langle 0|\rho_*|1\rangle|^2,$$
$$I_* = 1-\mathrm{tr}(\rho_*^2). \tag{16}$$

For $s_* = \langle 0|\rho_*|0\rangle$, the exact identity $C_* + I_*/2 = s_*(1-s_*)$ expresses the stationary population variance as the sum of these two contributions. The refined upper bound in Section V gives, uniformly as $\epsilon \downarrow 0$,

$$r_\epsilon \leq \frac{C_*}{\epsilon} + o(1), \qquad \frac{C_*}{\epsilon} + \frac{I_*}{2\epsilon} \leq 1 + o(1). \tag{17}$$

Indeed, $C_* = p + O(\epsilon^2)$ and $I_*/2 = \nu + O(\epsilon^2)$ under the stationary-state budget, while Eq. (41) gives $r_\epsilon \leq p/\epsilon + o(1)$. Thus a sequence with $r_\epsilon \to \eta > 0$ requires at least $\eta\epsilon$ of squared stationary coherence to leading order, and its impurity is at most $2(1-\eta)\epsilon + o(\epsilon)$. These constraints hold for every choice of stationary state. They are sharp: for each fixed $\eta \in [0,1]$, a mixed-state extension of the single-jump construction realizes $r_\epsilon \to \eta$, $C_*/\epsilon \to \eta$, and $I_*/(2\epsilon) \to 1-\eta$ at every sufficiently small actual disturbance; see Appendix C 4. They describe asymptotic sequences; the remainders must be retained when interpreting finite-error data.

The same stationary coherence limits full-state reset accuracy. For the actual finite channel $\Phi = e^{\mathcal{L}}$, define the worst-case trace-distance error over all qubit inputs,

$$E_{\mathrm{tr}}(\Phi) = \sup_\rho \tfrac{1}{2}\|\Phi(\rho) - |1\rangle\langle 1|\|_1. \tag{18}$$

A stationary state is itself an allowed input, so $E_{\mathrm{tr}}(\Phi) \geq \frac{1}{2}\|\rho_* - |1\rangle\langle 1|\|_1 \geq \sqrt{C_*}$. Hence Eq. (17) implies

$$\frac{E_{\mathrm{tr}}(\Phi)^2}{\epsilon} \geq r_\epsilon - o(1). \tag{19}$$

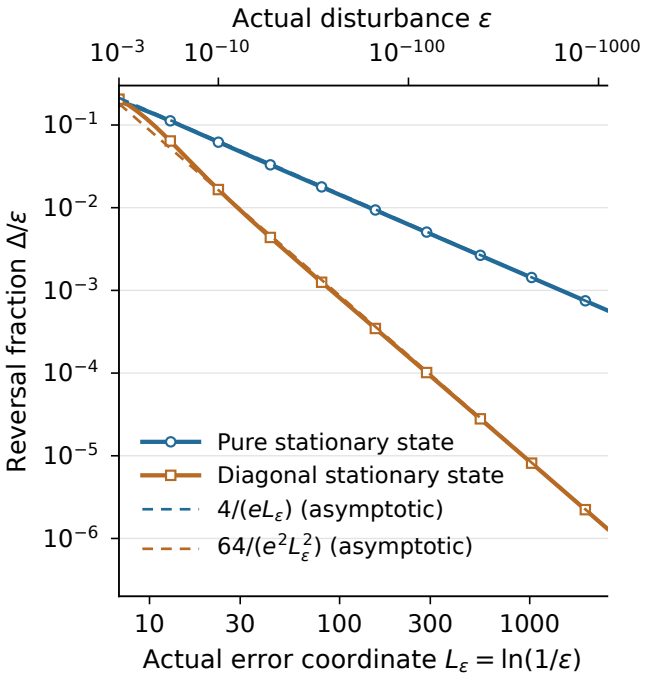


FIG. 1. Population-order reversal Δ/ϵ at matched actual disturbance $\epsilon = P(0|1)$, with $L_\epsilon = \log(1/\epsilon)$. Solid curves show the tilted pure-stationary-state construction (circles), Eqs. (21)–(23), and the diagonal-stationary-state construction (squares), Eqs. (C15)–(C18). Dashed curves give $4/(eL_\epsilon)$ and $64/(e^2L_\epsilon^2)$. The solid curves are attainable lower bounds. At $\epsilon = 10^{-3}$ the displayed diagonal construction exceeds the displayed pure-state construction; Theorems 1 and 2 describe the optimized gains as $\epsilon \to 0$.

For every fixed $0 < \eta \leq 1$, the matching construction also satisfies $E_{\mathrm{tr}}^2/\epsilon \to \eta$, making this tradeoff sharp to leading order. The consequences for the accuracy of population-optimal reset are discussed in Section VI.

IV. OPTIMAL CONSTRUCTIONS AND COHERENCE TRANSFER

The two bounds are attained through different population–coherence mechanisms. Figure 1 compares explicit realizations at the same actual disturbance ϵ. These curves give constructive lower bounds; their ordering at finite error can differ from that of the optimized classes.

A. Tilted pure stationary state

For a stationary eigenvector with overlap $p = |\langle 0|g\rangle|^2$, define the readout coefficients

$$v = 1-2p, \qquad w = 2\sqrt{p(1-p)},$$
$$V = v^2, \quad W = w^2, \quad R = vw. \tag{20}$$

Let $L \to \infty$ and define

$$p = e^{-L}, \qquad A = L + \log L + 1 - \log 4, \qquad B = A/L. \tag{21}$$

Choose orthonormal vectors

$$|g\rangle = \sqrt{p}|0\rangle + \sqrt{1-p}|1\rangle, \qquad |e\rangle = \sqrt{1-p}|0\rangle - \sqrt{p}|1\rangle, \tag{22}$$

and, in the ordered basis (e, g), take a single jump with no Hamiltonian:

$$J = \begin{pmatrix} -\sqrt{B} & 0 \\ \sqrt{A} & 0 \end{pmatrix}, \qquad H = 0. \tag{23}$$

For every sufficiently large finite L, this generator has stationary state $|g\rangle\langle g|$. For a state ρ, write $n = \langle e|\rho|e\rangle$ for its population and $z = \langle e|\rho|g\rangle$ for its coherence in the stationary eigenbasis. The measured population is $P_0 = p + vn + w \operatorname{Re} z$, and the dynamics obeys

$$\dot{n} = -An, \qquad \dot{z} = -\sqrt{AB}\, n - \Gamma z, \qquad \Gamma = (A+B)/2. \tag{24}$$

Set

$$F = e^{-A}, \qquad S = e^{-\Gamma}, \qquad J_0 = \int_0^1 e^{-At} e^{-\Gamma(1-t)}\, dt = \frac{S-F}{A-\Gamma}. \tag{25}$$

For the difference between the two input states, let u and z denote the population and coherence differences in these coordinates. Then $q = vu + w \operatorname{Re} z$, with $(u(0), z(0)) = (v, w)$, and direct solution gives

$$q = v^2 F + w^2 S - vw\sqrt{AB}\, J_0. \tag{26}$$

For the target input $|1\rangle$, the initial conditions are $n(0) = p$, $z(0) = -w/2$, giving the actual disturbance

$$\epsilon(L) = p + vpF - \frac{w^2}{2} S - pw\sqrt{AB}\, J_0. \tag{27}$$

In (26), the surviving population and coherence terms are positive, while population transfer into coherence produces the negative cross term. The tilt of the stationary eigenbasis makes this coherence directly visible in the population measurement. With $A \sim L$, $B/A = 1/L$, and $\epsilon \sim e^{-L}$, the negative cross term has twice the magnitude of the surviving population term to leading order, giving $\Delta \sim (4/e)\epsilon/\log(1/\epsilon)$. Appendix B 5 derives these asymptotics and shows that the family realizes every sufficiently small positive actual disturbance ϵ.

The representation (23) has $H = 0$, while the jump contains coherent amplitudes. Identity shifts of the jump redistribute terms between the dissipator and Hamiltonian, so the value of H alone does not characterize the coherent resources of the dynamics.

The same construction permits fixed jump strength. Since the squared operator norm is $\|J\|^2 = A + B$, any fixed rate $\kappa > 0$ gives the rescaling

$$J_{\text{phys}} = \sqrt{\frac{\kappa}{A+B}} J, \qquad t_{\text{phys}} = \frac{A+B}{\kappa} = \frac{\log(1/\epsilon) + \log\log(1/\epsilon) + O(1)}{\kappa}. \tag{28}$$

The rescaled dynamics gives the same channel with $\|J_{\text{phys}}\|^2 = \kappa$. The stationary-state direction and ratio B/A depend on the target error, while each chosen generator remains fixed during its evolution.

B. Diagonal stationary state

When $\rho_* = \operatorname{diag}(\nu, 1-\nu)$ in the measurement basis, stationarity gives

$$a = \nu + (1-\nu)q, \qquad \epsilon = \nu(1-q). \tag{29}$$

For $q = -\Delta < 0$, these relations become

$$\nu - a = (1-\nu)\Delta, \qquad \epsilon - \nu = \nu\Delta. \tag{30}$$

The unwanted population for input $|0\rangle$ transiently falls below its stationary value, while the target input ends slightly above that value. Classical two-state relaxation cannot cross the stationary population.

For the difference $\delta\rho$ between the two input trajectories, let $u = \delta\rho_{00}$ and $z = \delta\rho_{01}$ in the measurement basis. A single jump and a Hamiltonian can keep z real and realize

$$\frac{d}{dt}\begin{pmatrix} u \\ z \end{pmatrix} = \begin{pmatrix} -\alpha & r_0 \\ -c_0 & -\lambda_+ \end{pmatrix}\begin{pmatrix} u \\ z \end{pmatrix}, \qquad \lambda_+ = \frac{A(1+\chi)^2 + B}{2}, \tag{31}$$

where $\chi = \sqrt{\nu/(1-\nu)}$ and $\alpha, r_0, c_0, \lambda_+ > 0$. Appendices A and C specify the coefficients and the explicit GKSL realization. The input difference starts at $(u, z) = (1, 0)$, and the measured determinant is $q = u(1)$. Population first generates coherence through $-c_0 u$, which then feeds back into population through $r_0 z$. This negative feedback can transiently reverse the population difference even when stationary coherence is zero.

For both asymptotically optimal constructions, $\tau = B/A$ is of order $1/\log(1/\epsilon)$. The negative slow-mode amplitude is then $8\sqrt{\nu}\,\tau$ to leading order for the diagonal construction, compared with $4\sqrt{p\tau}$ for direct readout in the tilted pure-state construction. The competing positive population mode decays about twice as fast. Optimizing their difference gives, to leading order, one quarter of the squared slow-mode amplitude multiplied by $e^{-\tau L}$. The exponential factor comes from the $O(\tau)$ correction to the slow-to-fast decay-rate ratio $1/2$, with $L = \log(1/\nu)$ or $\log(1/p)$, respectively. The resulting leading gains are $16\nu\tau^2 e^{-\tau L}$ and $4p\tau e^{-\tau L}$, maximized at $\tau L = 2$ and $\tau L = 1$. The extra factor of τ accounts for the additional inverse logarithm in the diagonal case.

Appendix C proves the uniform upper bound for this class and gives a matching GKSL realization. In that family, $\nu - a \sim (64/e^2)\nu/\log^2(1/\nu)$.

Thus a diagonal stationary state permits transient coherence generated by $-c_0 u$. The two theorems quantify the distinct gains available from this feedback and from direct readout enabled by stationary coherence.

C. Input distinguishability at a population crossing

The two constructions illustrate how population order can reverse without an increase in quantum distinguishability. Write the difference between the two evolving inputs in the measurement basis as

$$\begin{aligned} \delta\rho(t) = \rho_0(t) - \rho_1(t) &= \begin{pmatrix} q(t) & c(t) \\ c(t)^* & -q(t) \end{pmatrix}, \\ D_{\mathrm{pop}}(t) &= |q(t)|, \\ D_{\mathrm{tr}}(t) = \tfrac{1}{2}\|\delta\rho(t)\|_1 &= \sqrt{q(t)^2 + |c(t)|^2}. \end{aligned} \tag{32}$$

The quantum trace distance D_{tr} is nonincreasing under the semigroup. At a finite population crossing t_c, $q(t_c) = 0$, yet $c(t_c) \neq 0$: the finite matrix exponential is invertible and cannot map the two distinct input states to the same state. All remaining input distinguishability is then in coherence. Subsequent population separation satisfies $|q(t)| \leq |c(t_c)|$ for $t \geq t_c$. The two logarithmic laws bound how much of this distinguishability can appear as reversed population order at a prescribed small disturbance. Figure 2 illustrates both mechanisms at the same actual readout disturbance $\epsilon_\star = 10^{-4}$. The upper panels resolve the population crossing, while the lower panels show the surviving quantum distinguishability.

A dephasing intervention isolates this mechanism. At t_c, apply the same channel $\mathcal{Z}_\lambda$ to both trajectories, leaving populations unchanged and multiplying coherences by $\lambda \in [0, 1]$, and then resume the original evolution. Since $\mathcal{Z}_\lambda[\delta\rho(t_c)] = \lambda\delta\rho(t_c)$, linearity gives exactly

$$q_\lambda(t) = \lambda q(t) \qquad (t \geq t_c). \tag{33}$$

Complete dephasing erases the later reversal. This intervention tests the coherence contribution; it is not part of the autonomous optimization defining G. The distinction between population and quantum distinguishability is the coherence-assisted mechanism studied in Refs. [3, 4].

V. UNIVERSAL UPPER BOUNDS

Optimality follows from an upper bound valid for arbitrary generators. A stationary-state constraint bounds the available error budget, and a population–coherence comparison bounds the transient reversal.

A. Stationary-state constraint

Every finite-dimensional GKSL generator has a stationary density matrix, obtained from a convergent subsequence of long-time orbit averages. For any fixed point ρ_* of the channel $\Phi = e^{\mathcal{L}}$, set $p_* = \langle 0|\rho_*|0\rangle$. The positive effect $\Phi^*(|0\rangle\langle 0|)$ has diagonal entries a, ϵ, so the magnitude of its off-diagonal entry is at most $\sqrt{a\epsilon}$. Positivity of ρ_* and stationarity give $p_* \leq (\sqrt{ap_*} + \sqrt{\epsilon(1-p_*)})^2$, and therefore

$$[(1 - \sqrt{a})^2 + \epsilon]p_* \leq \epsilon. \tag{34}$$

With (11), $p_* = \nu + (1 - 2\nu)p$. In the relevant region $a \leq \epsilon \downarrow 0$, the exact inversion in Appendix A yields

$$p + \nu \leq \varrho(\epsilon) = \epsilon[1 + O(\sqrt{\epsilon})]. \tag{35}$$

Stationary-state tilt and mixedness share the same small measured-error budget. Both must be retained to control the finer scale $\epsilon/\log(1/\epsilon)$.

B. Population–coherence comparison

In an arbitrary stationary eigenbasis, set $u = \langle e|\rho|e\rangle - \nu$ and $z = \langle e|\rho|g\rangle$. Their homogeneous equations have the form

$$\dot{u} = -\alpha u + \mathrm{Re}(R_0 z^*), \qquad \dot{z} = \widetilde{C}u + Ez^* - (\Gamma + i\omega)z. \tag{36}$$

Appendix A derives these equations for arbitrary jump collections and Hamiltonians and gives the exact coefficients in (A7)–(A8). The equations retain both population feedback and coupling to the conjugate coherence.

The frequency term $-i\omega z$ does not contribute directly to the derivative of $|z|$, making the comparison uniform in ω. Gram positivity bounds the coupling strengths by nonnegative constants r_0, c_0 and the coherence damping from below by λ. These bounds define the comparison matrix $K = \left(\begin{smallmatrix} -\alpha & r_0 \\ c_0 & -\lambda \end{smallmatrix}\right)$ with nonnegative off-diagonal entries and parameters given in (A11). Choose the relative phase of e, g so that the off-diagonal entry of $|0\rangle\langle 0|$ is nonnegative. The input difference obeys the same equations, with $(u(0), z(0)) = (v, w)$ and $q = vu + w\,\mathrm{Re}\,z$, using (20). Retaining the positive free decay of the population gives

$$q(t) \geq 2Ve^{-\alpha t} - (v, w)e^{tK}\begin{pmatrix} v \\ w \end{pmatrix} \qquad (t \geq 0). \tag{37}$$

The proof in Appendix A first bounds $|u|, |z|$ by a positive comparison system, then applies the variation-of-constants formula to the population. This step preserves the positive term competing with the negative coherence transient.

The rate parameters A, B are defined in (A4) from the jump entries in a stationary eigenbasis, after an identity shift sets the gg entries to zero. For the pure-state construction, they coincide with the rates in (23). At time

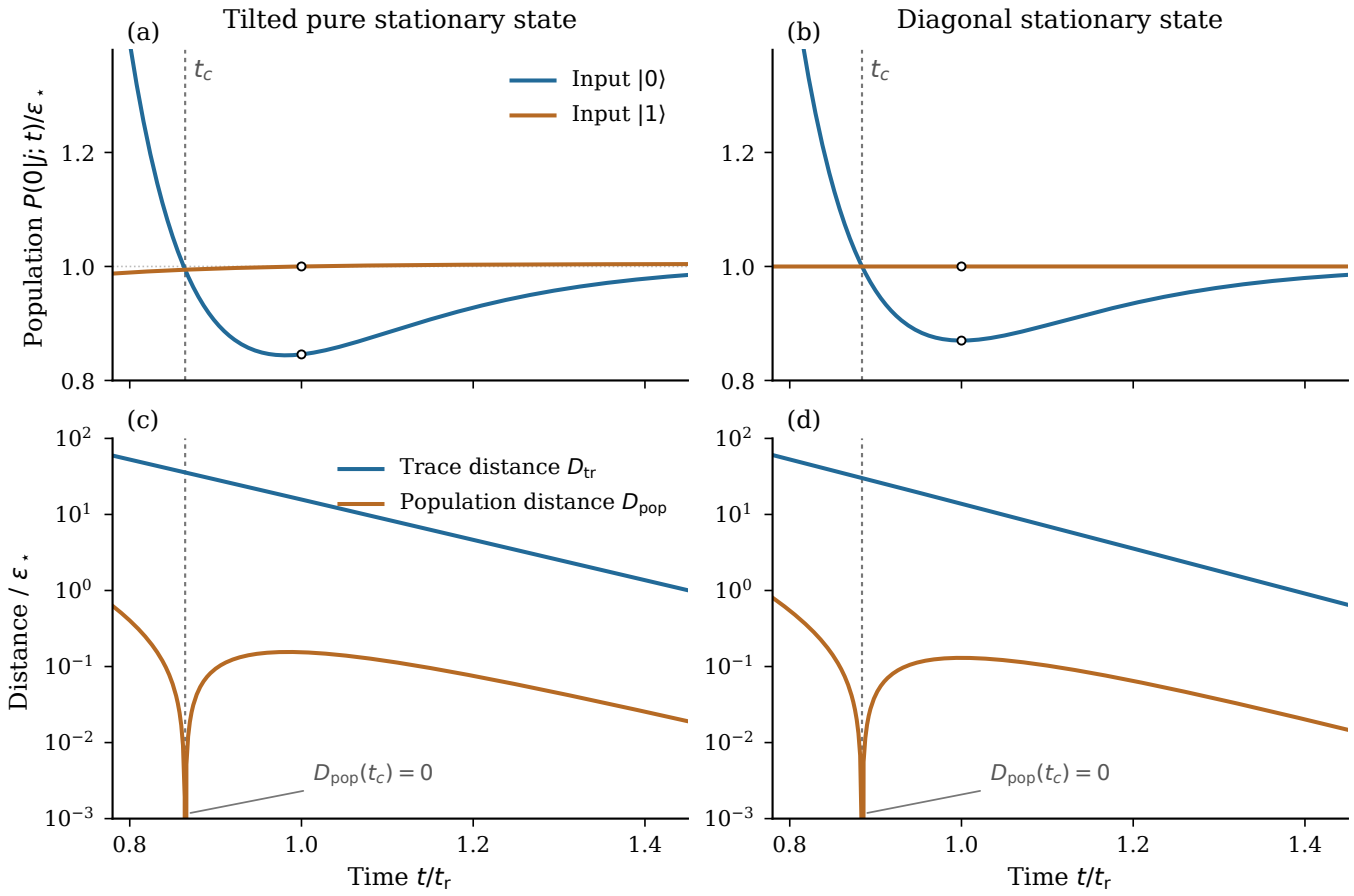


FIG. 2. Population reversal and remaining input distinguishability in the two explicit constructions. Both generators give the same actual disturbance $\epsilon_\star = P(0|1;t_{\mathrm{r}}) = 10^{-4}$ at readout time $t_{\mathrm{r}} = 1$. (a),(b) The input-conditioned populations cross at t_c (vertical dashed lines) and subsequently separate with reversed order. Open circles mark $t = t_{\mathrm{r}}$. (c),(d) The population distance $D_{\mathrm{pop}} = |q|$ vanishes at the crossing and then increases, while the quantum trace distance D_{tr} decreases. At the crossing, the nonzero quantum distance is entirely due to the off-diagonal difference. Populations and distances are normalized by $\epsilon_\star$; zero on the logarithmic lower axes is indicated by annotation. The curves are finite realizations evaluated from the exact two-dimensional evolution and checked against the full GKSL propagator, using double-precision arithmetic.

one, define

$$\mathcal{E}(p,\nu,A,B) = \left[(v,w)e^{K}\begin{pmatrix} v \\ w \end{pmatrix} - 2Ve^{-\alpha}\right]_+, \qquad [x]_+ = \max(x,0). \tag{38}$$

Then $(\epsilon - a)_+ \le \mathcal{E}$. The positive comparison system enlarges the physical class by bounding its couplings independently.

Proposition 4 (Sharp uniform envelope). Let $p, \nu \ge 0$, $p + \nu \le \rho$, and $A, B \ge 0$. With the definitions above, as $\rho \downarrow 0$,

$$\mathcal{E}(p,\nu,A,B) \le \frac{4p/e + o(\rho)}{\log(1/\rho)}. \tag{39}$$

The remainder is a function of ρ alone and is uniform in all four parameters.

The leading contribution comes from a small window of the rate ratio $\tau = B/A$. With $L = \log(1/\rho)$, optimizing the overall rate and using the budget constraint $p+\nu \le \rho$ give

$$\mathcal{E} \le [4p + o(\rho)]\tau e^{-\tau L} \le \frac{4p/e + o(\rho)}{L}. \tag{40}$$

The factor $\tau e^{-\tau L}$ is maximized at $\tau L = 1$. Appendix B establishes this estimate uniformly and bounds the remaining rate ratios, including zero rates, repeated eigenvalues, and arbitrarily large rates. Their contributions are $o(\rho/L)$.

C. Sharp asymptotics and saturation

For a finite realization with $a \le \epsilon$, choose any stationary state. Lemma 5 gives $p + \nu \le \varrho(\epsilon)$, and Lemma 6 and Proposition 4 then yield

$$\epsilon - a \le \frac{4p/e + o(\varrho(\epsilon))}{\log(1/\varrho(\epsilon))} \le \left(\frac{4}{e} + o(1)\right)\frac{\epsilon}{\log(1/\epsilon)}. \tag{41}$$

The last step uses $\varrho(\epsilon) = \epsilon[1+O(\sqrt{\epsilon})]$. If $a > \epsilon$, the upper bound is immediate.

Uniformity means that, for every $\delta > 0$, there is an $\epsilon_\delta > 0$ such that every finite generator obeys

$$\epsilon - a \leq \left(\frac{4}{e} + \delta\right) \frac{\epsilon}{\log(1/\epsilon)} \qquad (0 < \epsilon < \epsilon_\delta). \tag{42}$$

Fix a point in the probability closure with $0 < \epsilon < \epsilon_\delta$ and approximate it by finite realizations. The approximants eventually lie in the same error interval. Continuity of the right-hand side in ϵ then extends (42) to $\mathcal{Q}_2$.

The pure-state construction in Section IV supplies the matching lower bound, with its actual disturbance and coverage established in Appendix B 5, completing Theorem 1. The refined coefficient $4p/e$ also proves Corollary 3:

Proof. For all sufficiently large n, $a_n < \epsilon_n$. The budget gives $p_n + \nu_n \leq \epsilon_n[1 + o(1)]$. The refined first inequality of (41), multiplied by $\log(1/\epsilon_n)/\epsilon_n$, and (12) force $\liminf p_n/\epsilon_n \geq 1$. Both limits follow from nonnegativity and the budget. □

The coherence and purity identities in (14) follow directly from the stationary eigenvalues and the overlap p_n.

For the diagonal class, $p = 0$ leaves only feedback from population through coherence and back into population. Appendix C refines the comparison to the scale $\nu/\log^2(1/\nu)$, proves a uniform bound over all rate ratios, and constructs a matching GKSL generator. Equation (29) converts ν to the actual disturbance ϵ, and the same probability-closure argument completes Theorem 2.

VI. DISCUSSION

The endpoint laws turn conditional population measurements into constraints on stationary coherence. Equation (17) quantifies the minimum coherent fraction required by a given fraction of the leading reversal. The diagonal-stationary-state law describes the smaller reversal available through transient coherence alone. At full leading-order saturation, every stationary state must become nearly pure, with $C_* \sim \epsilon$. Stationary coherence vanishes in this limit, but remains necessary to attain the larger of the two asymptotic scales.

The full-state tradeoff in Eq. (19) follows because reset must also act on its own stationary state. More explicitly,

$$E_{\mathrm{tr}}(\Phi) \geq \tfrac{1}{2}\|\rho_* - |1\rangle\langle 1|\|_1 = \sqrt{\nu^2 + (1-2\nu)p} \geq \sqrt{C_*},$$
$$E_{\mathrm{tr}}(\Phi) \leq \sqrt{a+\epsilon}. \tag{43}$$

For the upper bound, positivity of $\Phi^*(|0\rangle\langle 0|)$ bounds every output population by its trace $a + \epsilon$, and the trace distance to a pure target is at most the square root of the unwanted population. The stationary-state lower bound and Eq. (17) give Eq. (19). Together with the upper bound, they imply that every sequence saturating the unrestricted population law obeys

$$[1 - o(1)]\sqrt{\epsilon} \leq E_{\mathrm{tr}}(\Phi) \leq \sqrt{2\epsilon}. \tag{44}$$

The pure-state construction attains the lower leading coefficient, $E_{\mathrm{tr}}(\Phi) \sim \sqrt{\epsilon}$, as shown in Appendix B 5. In comparison, incoherent qubit relaxation can approach the replacement channel $\rho \mapsto \mathrm{diag}(\epsilon, 1-\epsilon)$ at the same actual disturbance; that channel has $\Delta = 0$ and $E_{\mathrm{tr}} = \epsilon$. Thus maximizing the conditional population reversal requires a parametrically larger full-state error than this incoherent limit. The residual stationary coherence accounts for the difference. Population readout alone can therefore reward dynamics that retain appreciably more coherent error relative to the target state. An additional final dephasing step preserves both measured probabilities and reduces the worst-case trace-distance error to at most $a + \epsilon \leq 2\epsilon$. This added operation lies outside the fixed-generator reset task, identifying autonomy as the origin of the larger full-state error.

Autonomy imposes a separate restriction. Time-dependent control can first damp toward $|0\rangle$, leaving population ϵ in $|1\rangle$ for input $|1\rangle$, and then apply a unitary swap. This gives $a = 0$ and $P(0|1) = \epsilon$, hence $G_{\mathrm{td}}(\epsilon) = \epsilon$. With the same system dimension and measurement basis, a fixed generator reaches only a vanishing fraction of this reversal. The comparisons concern attainable probabilities; no bound on time or coupling strength is imposed.

Both attaining families are contained in the earlier four-parameter ansatz or its probability closure. That ansatz uses $H = X$, where X is the Pauli x matrix, and a rank-one jump between states of the form $(\cos\theta, i\sin\theta)^T$ [1]. The pure-state family lies in its probability closure, while the diagonal-state family belongs to the ansatz itself at sufficiently small error. Appendix D gives the transformations. The upper bounds establish their endpoint optimality. The earlier exclusion bound implies $a \geq (\epsilon/2)^{10}[1 + o(1)]$ near reset, whereas Theorem 1 determines $a/\epsilon \to 1$ and its leading correction. Appendix E exhibits the additional excluded region $\epsilon/2 < a < 3\epsilon/4$ for sufficiently small ϵ.

The finite-error optimal curve and quantitative finite-error versions of the coherence constraint remain open. Such estimates would turn the asymptotic structural inference into bounds usable at specified experimental errors.

VII. CONCLUSIONS

Time-independent qubit dynamics permits a sharply limited reversal of population order near reset. Its optimal scale is $\epsilon/\log(1/\epsilon)$, with coefficient $4/e$; admitting a diagonal stationary state reduces it to $\epsilon/\log^2(1/\epsilon)$, with coefficient $64/e^2$. These laws also constrain the coherence and purity of every stationary state. A finite fraction of the leading population reversal requires stationary coherence, and full saturation entails a worst-case trace-distance reset error of order $\sqrt{\epsilon}$. The sharp population boundary thus

identifies both the coherent structure needed to attain it and its cost in full-state accuracy.

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Appendix A: Stationary-state reduction

This appendix derives the stationary error budget and the population–coherence comparison used in Section V.

1. Stationary-state constraint

Every finite-dimensional GKSL generator has a stationary density matrix. Compactness gives a convergent subsequence of the time averages of any orbit, and $\mathcal{L}(t^{-1}\int_0^t e^{s\mathcal{L}}(\rho)\,ds) = (e^{t\mathcal{L}}(\rho)-\rho)/t \to 0$. The following lemma bounds every such state in terms of the two measured errors.

Lemma 5 (Stationary error budget). Let Φ be a qubit channel with the probabilities in (1). For any fixed point ρ_* of Φ, write $p_* = \langle 0|\rho_*|0\rangle$. Then the inequality (34) holds. If $a \le \epsilon < 1/4$, diagonalize ρ_* as in (11). Then $p, \nu < 1/2$ and

$$p_* = \nu + (1-2\nu)p, \tag{A1}$$

$$p+\nu \le \varrho(\epsilon) := 1-\sqrt{1-2b_*(\epsilon)},$$
$$b_*(\epsilon) = \frac{\epsilon}{1-2\sqrt{\epsilon}+2\epsilon}, \tag{A2}$$
$$\varrho(\epsilon) = \epsilon[1+O(\sqrt{\epsilon})].$$

Proof. The positive effect $F = \Phi^*(|0\rangle\langle 0|)$ has diagonal entries a, ϵ, so $|F_{01}| \le \sqrt{a\epsilon}$. Positivity of ρ_* similarly bounds its off-diagonal entry. Since $\Phi(\rho_*) = \rho_*$,

$$p_* = \mathrm{tr}(F\rho_*) \le \left(\sqrt{ap_*} + \sqrt{\epsilon(1-p_*)}\right)^2.$$

Taking square roots and rearranging proves (34). For $a \le \epsilon < 1/4$, it gives $p_* \le b_*(\epsilon) < 1/2$. Equation (A1) implies $\nu \le p_*$ and $p \le p_*$, so $s = p+\nu < 1$. Since

$$p_* = s - 2\nu p \ge s - \frac{s^2}{2},$$

and the right-hand side increases on $[0,1)$, inversion yields (A2). □

2. Population–coherence dynamics

Fix an arbitrary stationary state and use the eigenbasis (e, g) from (11). Subtracting each jump's gg entry times the identity, with the corresponding Hamiltonian adjustment, preserves the generator and gives

$$J_\mu = \begin{pmatrix} x_\mu & y_\mu \\ z_\mu & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} \omega/2 & h \\ h^* & -\omega/2 \end{pmatrix}. \tag{A3}$$

An irrelevant scalar multiple of the identity has also been removed from H. Define

$$A = \sum_\mu |z_\mu|^2, \quad B = \sum_\mu |x_\mu|^2, \quad U = \sum_\mu |y_\mu|^2,$$
$$C = \sum_\mu x_\mu z_\mu^*, \quad D = \sum_\mu x_\mu^* y_\mu, \quad E = \sum_\mu y_\mu z_\mu^*. \tag{A4}$$

These coefficients form the positive semidefinite Gram matrix

$$\begin{pmatrix} B & D^* & C \\ D & U & E \\ C^* & E^* & A \end{pmatrix} \succeq 0, \tag{A5}$$

whose principal minors imply $|C| \le \sqrt{AB}$, $|D| \le \sqrt{BU}$, and $|E| \le \sqrt{AU}$. Stationarity gives the exact relations

$$U = \frac{A\nu}{1-\nu}, \qquad h = -i\frac{\nu C - D/2}{1-2\nu}. \tag{A6}$$

The real frequency ω remains unrestricted.

Let $u = \langle e|\rho|e\rangle - \nu$ be the $|e\rangle$ population relative to stationarity and $z = \langle e|\rho|g\rangle$ the coherence. Direct substitution in (4) gives

$$\dot{u} = -(A+U)u + \mathrm{Re}(R_0 z^*), \tag{A7}$$

$$\dot{z} = \widetilde{C}u + Ez^* - (\Gamma + i\omega)z, \tag{A8}$$

$$\widetilde{C} = \frac{C-D}{1-2\nu},$$
$$R_0 = \frac{2[(1-\nu)D - \nu C]}{1-2\nu},$$
$$\Gamma = \frac{A+B+U}{2}.$$

The same homogeneous equations govern the difference between two trajectories, including the population-feedback term in (A7) and the conjugate-coherence term in (A8). For $\nu = 0$, stationarity forces $U = D = E = R_0 = h = 0$, reducing the dynamics to a triangular population–coherence system.

Choose the relative phase of e, g so that the off-diagonal entry of $|0\rangle\langle 0|$ is nonnegative. Set v, w, V, W, R as in (20).

In particular, $V + W = 1$. The difference between the two preparation-basis inputs is

$$|0\rangle\langle 0| - |1\rangle\langle 1| = \begin{pmatrix} v & w \\ w & -v \end{pmatrix}, \qquad (u(0), z(0)) = (v, w), \tag{A9}$$

and the measured determinant at time t is

$$q(t) = vu(t) + w \operatorname{Re} z(t). \tag{A10}$$

The basis change therefore transforms the preparation and measurement axes together.

3. Positive comparison

Introduce

$$\chi = \sqrt{\frac{\nu}{1-\nu}}, \quad \alpha = A(1+\chi^2), \quad \lambda = \frac{A(1-\chi)^2 + B}{2},$$
$$c_0 = \frac{\sqrt{AB}(1+\chi^2)}{1-\chi}, \quad r_0 = \frac{2\chi\sqrt{AB}}{1-\chi},$$
$$K = \begin{pmatrix} -\alpha & r_0 \\ c_0 & -\lambda \end{pmatrix}. \tag{A11}$$

The Gram inequalities and (A6) imply

$$\Gamma - |E| \geq \lambda, \qquad |\widetilde{C}| \leq c_0, \qquad |R_0| \leq r_0. \tag{A12}$$

These bounds use necessary consequences of Gram positivity. Their extremal values need not be simultaneously attainable by a physical generator.

Lemma 6 (Positive comparison). For $0 \leq p \leq 1/2$ and $0 \leq \nu < 1/2$, every evolution above satisfies (37). This includes zero rates and arbitrary ω.

Proof. Equations (A7)–(A12) give upper right Dini-derivative bounds

$$D^+|u| \leq -\alpha|u| + r_0|z|, \qquad D^+|z| \leq c_0|u| - \lambda|z|.$$

The term $-i\omega z$ contributes zero to the derivative of $|z|$. At a zero of u or z, the corresponding modulus derivative gives the same inequality. Since K has nonnegative off-diagonal entries, its exponential has nonnegative entries. The componentwise comparison principle therefore gives $|u| \leq X$ and $|z| \leq Z$ for $(X, Z)^T = e^{tK}(v, w)^T$. This principle follows by adding a strictly positive perturbation to the comparison initial data and derivative, excluding a first contact, and then removing the perturbation.

The variation-of-constants formula for the population gives the stronger lower bound

$$u(t) \geq ve^{-\alpha t} - r_0 \int_0^t e^{-\alpha(t-s)} Z(s)\, ds$$
$$= 2ve^{-\alpha t} - X(t).$$

Substitute this and $\operatorname{Re} z \geq -Z$ into (A10); $v, w \geq 0$ yields (37). □

The free population decay in (37) supplies the positive term that competes with the coherence transient; replacing u by $-|u|$ would discard this term. At time one, the comparison gives the envelope (38), with $(\epsilon - a)_+ \leq \mathcal{E}$.

Appendix B: Uniform envelope bound

The proof of Proposition 4 separates four ranges of the rate ratio. Throughout this appendix, C_{b} denotes a finite universal constant that may increase between estimates. All thresholds depend only on ρ.

1. Dimensionless formulation

If $A = 0$, then $r_0 = c_0 = \alpha = 0$ and $\mathcal{E} = [We^{-B/2} - V]_+ = 0$ for small ρ. For $A > 0$, write

$$\tau = B/A, \qquad \mathfrak{a} = 1 + \chi^2, \qquad \ell = \frac{(1-\chi)^2 + \tau}{2},$$
$$c = \frac{\sqrt{\tau}(1+\chi^2)}{1-\chi}, \quad r = \frac{2\chi\sqrt{\tau}}{1-\chi}, \quad h_0 = rc, \tag{B1}$$

so that $K = AM$ with $M = \left(\begin{smallmatrix} -\mathfrak{a} & r \\ c & -\ell \end{smallmatrix}\right)$. Its decay rates are

$$\beta = \frac{\mathfrak{a} + \ell - d}{2}, \qquad \gamma = \frac{\mathfrak{a} + \ell + d}{2},$$
$$d = \sqrt{(\mathfrak{a} - \ell)^2 + 4h_0}. \tag{B2}$$

Since

$$\frac{h_0}{\mathfrak{a}\ell} = \frac{4\chi\tau}{(1-\chi)^2[(1-\chi)^2 + \tau]} \leq \kappa := \frac{4\chi}{(1-\chi)^2} = o(1), \tag{B3}$$

the roots satisfy $0 < \beta \leq \min(\mathfrak{a}, \ell)$ and $\gamma \geq \max(\mathfrak{a}, \ell)$ for small ρ. The estimates also use

$$\beta \geq (1 - \sqrt{\kappa}) \min(\mathfrak{a}, \ell). \tag{B4}$$

To see this, set $t = (1 - \sqrt{\kappa}) \min(\mathfrak{a}, \ell)$. Both $\mathfrak{a} - t \geq \sqrt{\kappa}\,\mathfrak{a}$ and $\ell - t \geq \sqrt{\kappa}\,\ell$, so $(\mathfrak{a} - t)(\ell - t) \geq h_0$; the smaller root of this characteristic equation is β.

For $0 \leq \tau \leq 1/2$, the spectral separation satisfies

$$g := \mathfrak{a} - \ell = \frac{(1+\chi)^2 - \tau}{2} \geq \frac{1}{4}. \tag{B5}$$

Define the nonnegative slow-mode weight

$$Q = \frac{(\gamma - \mathfrak{a})V + (\gamma - \ell)W + (r + c)R}{d}. \tag{B6}$$

The exact identity

$$(v, w)e^{AM}\begin{pmatrix} v \\ w \end{pmatrix} = Qe^{-\beta A} + (1 - Q)e^{-\gamma A} \tag{B7}$$

and $\gamma - \mathfrak{a} = h_0/[(d+g)/2] \leq h_0/g$ imply

$$Q = W + \frac{(\gamma - \mathfrak{a})(V - W) + (r + c)R}{d}$$
$$\leq 4p + \frac{h_0}{g^2} + \frac{2\sqrt{p}(r + c)}{g} \leq C_{\mathrm{b}}[p + \chi\tau + \sqrt{p\tau}]. \tag{B8}$$

In particular $Q \le C_{\rm b}\sqrt{\rho} < 1$. Since $\gamma \ge \mathfrak{a}$,

$$\mathcal{E} \le [Qe^{-\beta A} - Pe^{-\mathfrak{a}A}]_+, \qquad P = V - W + Q = 1 + O(\sqrt{\rho}). \tag{B9}$$

Put $\theta = \beta/\mathfrak{a}$. Equations (B5) and (B2) give

$$\frac{1+\tau}{2} - C_{\rm b}\chi \le \theta \le \frac{1+\tau}{2}. \tag{B10}$$

For the upper inequality one can use $\ell/\mathfrak{a} = (1+\tau)/2 - \chi/(1+\chi^2) - \tau\chi^2/[2(1+\chi^2)]$. The lower inequality follows by subtracting $(\gamma - \mathfrak{a})/\mathfrak{a} = O(\chi\tau)$.

For $Q > 0$, the maximum of $Qx^\theta - Px$ over $x = e^{-\mathfrak{a}A} \in [0,1]$ occurs in the interior because $\theta Q < P$. Its exact value is

$$D_{\max} = (1-\theta)\theta^{\theta/(1-\theta)} Q^{1/(1-\theta)} P^{-\theta/(1-\theta)}. \tag{B11}$$

For $Q = 0$, the envelope vanishes. Equation (B11) optimizes over all overall rates A, including unbounded-rate sequences.

2. Large rate ratios

For $\tau \ge 1/2$, small ρ ensures

$$\beta \ge 7/10, \qquad 1 \le \mathfrak{a} \le 101/100, \tag{B12}$$

by (B4). To obtain a bound uniform near repeated rates without dividing by d, introduce the positive convolution

$$J(s) = \int_0^s e^{-\beta(s-t)} e^{-\gamma t}\, dt \le s e^{-\beta s}. \tag{B13}$$

When $d > 0$, also $J(s) \le d^{-1} e^{-\beta s}$. Matrix exponentiation or variation of constants gives

$$(e^{sM})_{12} = rJ(s), \qquad (e^{sM})_{21} = cJ(s), \tag{B14}$$

$$(e^{sM})_{11} - e^{-\mathfrak{a}s} = h_0 \int_0^s e^{-\mathfrak{a}(s-t)} J(t)\, dt \ge 0, \tag{B15}$$

$$0 \le (e^{sM})_{22} \le e^{-\beta s}. \tag{B16}$$

The last inequality follows from the two nonnegative spectral weights, whose sum is one, and extends continuously to $d = 0$.

On $1/2 \le \tau \le 4$, the coefficients obey $h_0 \le C_{\rm b}\chi$ and $r + c \le C_{\rm b}$. Equations (B12)–(B15) give respective bounds $C_{\rm b}\chi s^2 e^{-7s/10}$ and $C_{\rm b} s e^{-7s/10}$ for the correction in (B15) and for $(r+c)J(s)$. On $\tau \ge 4$, small ρ gives $d \ge \ell - \mathfrak{a} \ge \tau/4$, while $h_0 \le C_{\rm b}\chi\tau$ and $r + c \le C_{\rm b}\sqrt{\tau}$. The corresponding bounds are $C_{\rm b}\chi s e^{-7s/10}$ and $C_{\rm b}\tau^{-1/2} e^{-7s/10}$. Absorbing the polynomial factors therefore gives, uniformly on the entire large-ratio region,

$$\begin{aligned} (e^{sM})_{11} &\le e^{-\mathfrak{a}s} + C_{\rm b}\chi e^{-3s/5}, \\ (r+c)J(s) &\le C_{\rm b} e^{-3s/5}, \qquad (e^{sM})_{22} \le e^{-7s/10}. \end{aligned} \tag{B17}$$

Consequently,

$$\mathcal{E} \le [-Ve^{-\mathfrak{a}A} + C_{\rm b}\sqrt{\rho}\, e^{-3A/5}]_+. \tag{B18}$$

Optimize over $x = e^{-\mathfrak{a}A}$, with exponent $t = 3/(5\mathfrak{a})$. The maximum is at most $C_{\rm b}\rho^{1/[2(1-t)]}$. Since $\mathfrak{a} \le 101/100$, this exponent is at least $101/82 > 6/5$. Hence

$$\sup_{\tau \ge 1/2, A \ge 0} \mathcal{E} \le C_{\rm b}\rho^{6/5} = o\big(\rho/\log(1/\rho)\big). \tag{B19}$$

This estimate includes coincident decay rates and arbitrarily large B/A.

3. Intermediate and small rate ratios

Set

$$L = \log(1/\rho), \qquad \eta = L^{-1/2}. \tag{B20}$$

For $\eta \le \tau \le 1/2$, Eqs. (B8) and (B10) yield $Q \le C_{\rm b}\sqrt{\rho}$ and $1/(1-\theta) \ge 2/(1 - \tau + C_{\rm b}\sqrt{\rho})$. All remaining factors in (B11) are uniformly bounded. Thus

$$D_{\max} \le C_{\rm b}(C_{\rm b}\sqrt{\rho})^{2/(1-\tau+C_{\rm b}\sqrt{\rho})} \le C_{\rm b}\rho e^{-\sqrt{L}/2}. \tag{B21}$$

For the last step, the logarithm after division by ρ is at most $C_{\rm b} - L(\tau - C_{\rm b}\sqrt{\rho})/(1 - \tau + C_{\rm b}\sqrt{\rho}) \le C_{\rm b} - \sqrt{L}/2$.

On $0 \le \tau \le \rho^{1/4}$, including $B = 0$, the finer form of (B8) gives $Q \le C_{\rm b}\rho^{5/8}$. Since $1/(1-\theta) \ge 2 - C_{\rm b}\sqrt{\rho}$,

$$D_{\max} \le C_{\rm b}\rho^{5/4 - C_{\rm b}\sqrt{\rho}} \le C_{\rm b}\rho^{6/5}. \tag{B22}$$

This estimate also permits $\theta < 1/2$, which is possible for mixed stationary states.

4. Leading asymptotics

It remains to treat

$$\rho^{1/4} \le \tau \le L^{-1/2}. \tag{B23}$$

Here $\chi \le C_{\rm b}\sqrt{\rho} = o(\tau)$. The two coupling terms in (B8) admit the sharper estimates

$$\begin{aligned} \frac{h_0}{g^2} &= \frac{8\chi(1+\chi^2)\tau}{(1-\chi)^2[(1+\chi)^2 - \tau]^2} \\ &\le 8\sqrt{\nu}\,\tau[1 + O(\eta + \sqrt{\rho})], \\ \frac{2\sqrt{p}(r+c)}{g} &= \frac{4\sqrt{p\tau}(1+\chi)^2}{(1-\chi)[(1+\chi)^2 - \tau]} \\ &\le 4\sqrt{p\tau}[1 + O(\eta + \sqrt{\rho})]. \end{aligned} \tag{B24}$$

Therefore

$$Q \le 4p + [1 + O(\eta + \sqrt{\rho})](4\sqrt{p\tau} + 8\sqrt{\nu}\,\tau) \le C_{\rm b}\sqrt{\rho\tau}. \tag{B25}$$

Squaring this bound while retaining the contribution from stationary-state tilt gives

$$\frac{Q^2}{4} \le 4p\tau + o(\rho\tau) \tag{B26}$$

uniformly. The remaining square and cross terms are bounded by constant multiples of $\rho\tau\sqrt{\eta}$, $\rho^{3/2}\sqrt{\tau}$, and ρ^2. After division by $\rho\tau$, these tend to zero uniformly by (B23); the relative error in the retained term is $O(\eta+\sqrt{\rho})$.

Write $m = 1/(1-\theta)$. Equation (B10) gives

$$m - 2 \geq 2\tau - C_{\mathrm{b}}\chi > 0, \qquad m = 2 + o(1). \tag{B27}$$

For $Q > 0$, use $Q \leq Q_0 = C_{\mathrm{b}}\sqrt{\rho\tau} < 1$ to obtain

$$\begin{aligned} Q^{m-2} &\leq Q_0^{m-2} \leq \exp[(2\tau - C_{\mathrm{b}}\chi)\log Q_0] \\ &\leq [1 + o(1)]e^{-\tau L}. \end{aligned} \tag{B28}$$

Indeed, $\log Q_0 = -L/2 + (\log\tau)/2 + \log C_{\mathrm{b}}$ and $|\log\tau| \leq L/4$, so the excess exponent above $-\tau L$ is at most $C_{\mathrm{b}}\eta + C_{\mathrm{b}}\sqrt{\rho}L = o(1)$. The prefactor in (B11) satisfies

$$(1-\theta)\theta^{\theta/(1-\theta)}P^{-\theta/(1-\theta)} = \tfrac{1}{4}[1 + o(1)]. \tag{B29}$$

Equations (B26)–(B29) now imply

$$\mathcal{E} \leq D_{\max} \leq [4p + o(\rho)]\tau e^{-\tau L} \leq \frac{4p/e + o(\rho)}{L}, \tag{B30}$$

using $\sup_{\tau\geq 0}\tau e^{-\tau L} = 1/(eL)$. Equations (B19), (B21), (B22), and (B30) together prove Proposition 4. Every remainder is uniform in the rates, their ratio, and the allocation of the stationary error budget. □

5. Matching construction at fixed error

The generator, kernels, and measured probabilities are given in (21)–(27). Their asymptotics establish the matching lower bound at fixed actual disturbance.

The parameter choice (21) gives

$$\begin{aligned} F = \frac{4p}{eL}, \qquad S = \left(\frac{2}{e} + o(1)\right)\sqrt{\frac{p}{L}}, \\ \sqrt{AB}\,J_0 = \frac{2/\sqrt{L}}{1 - 1/L}(S - F). \end{aligned} \tag{B31}$$

Hence the negative cross term in (26) is $(8/e + o(1))p/L$, while its positive population term is $(4/e + o(1))p/L$ and $w^2 S = o(p/L)$. Thus

$$\begin{aligned} q &= -\left(\frac{4}{e} + o(1)\right)\frac{p}{L}, \\ \epsilon(L) &= p[1 + o(1)], \\ \log\frac{1}{\epsilon(L)} &= L + o(1). \end{aligned} \tag{B32}$$

Thus the construction attains the leading coefficient in Theorem 1 at its actual measured disturbance.

The construction realizes every sufficiently small positive disturbance. The function $\epsilon(L)$ is continuous, positive for all sufficiently large L, and tends to zero. Its range on a connected tail therefore contains an interval $(0, \epsilon_0]$. Any sequence of parameter values realizing disturbances that tend to zero must have $L \to \infty$, since $\epsilon(L)$ has a positive minimum on each compact subinterval of that tail. The lower asymptotic bound therefore holds throughout a punctured neighborhood of zero. Together with (41), this completes the proof of Theorem 1.

The full-state error of this family follows from the same solution. For an arbitrary input, write $n = \langle e|\rho|e\rangle \in [0,1]$ and $z = \langle e|\rho|g\rangle$, with $|z| \leq 1/2$. At time one, $n' = Fn$ and $z' = Sz - kn$, where $k = \sqrt{AB}J_0$. Since the stationary state is $|g\rangle\langle g|$,

$$\begin{aligned} \sup_{\rho} \tfrac{1}{2}\|\Phi(\rho) - |g\rangle\langle g|\|_1 &\leq F + S/2 + k \\ &= O\Big(\sqrt{p/L}\Big). \end{aligned} \tag{B33}$$

The stationary input gives $E_{\mathrm{tr}} \geq \frac{1}{2}\||g\rangle\langle g| - |1\rangle\langle 1|\|_1 = \sqrt{p}$, while the triangle inequality gives the upper bound $\sqrt{p} + O(\sqrt{p/L})$. Hence $E_{\mathrm{tr}} \sim \sqrt{p} \sim \sqrt{\epsilon}$. This is an error estimate for the same finite generators that attain the leading population-reversal law.

Appendix C: Diagonal stationary states

This appendix proves Theorem 2 using the full mixed-state equations (A7)–(A8). For a stationary state diagonal in the measurement basis, the population ν satisfies $\nu = a\nu + \epsilon(1-\nu)$, which gives (29). Only $q < 0$ contributes to the reversal, and then $0 < \nu \leq \epsilon$. For small ϵ, the stationary state has $\nu < 1/2$, $|e\rangle = |0\rangle$, $|g\rangle = |1\rangle$, and $p = 0$. The input difference starts at $(u, z) = (1, 0)$ and gives $q = u(1)$. If $A = 0$ or $B = 0$, the Gram constraints decouple population from coherence, giving $q = e^{-\alpha} \geq 0$. It therefore suffices to consider $A, B > 0$.

1. Feedback bound

Let $(X, Z)^T = e^{tK}(1, 0)^T$. Lemma 6 gives $u(t) \geq 2e^{-\alpha t} - X(t)$. Introduce dimensionless time $x = \alpha t$ and the parameters

$$\begin{aligned} \tau = \frac{B}{A}, \qquad \zeta = \frac{\lambda}{\alpha} = \frac{(1-\chi)^2 + \tau}{2(1+\chi^2)}, \\ h_f = \frac{r_0 c_0}{\alpha^2} = \frac{2\chi\tau}{(1-\chi)^2(1+\chi^2)}. \end{aligned} \tag{C1}$$

With $F(x) = e^{-x}$ and convolution on the positive half-line, the comparison population satisfies

$$\begin{aligned} X &= F + h_f(F * e^{-\zeta\,\cdot} * X), \\ q &\geq 2F(x) - X(x) \quad \text{at } x = \alpha. \end{aligned} \tag{C2}$$

Varying A covers every $x \geq 0$ at fixed ν, τ.

For $\tau \geq 1/2$ and $\chi \leq 1/100$, put $k = 2/3$. Direct substitution gives

$$\begin{aligned}\eta_f &:= \frac{h_f}{(1-k)(\zeta-k)} \\ &= \frac{36\chi\tau}{(1-\chi)^2(3\tau-1-6\chi-\chi^2)} \\ &\leq 48\chi < 1.\end{aligned} \tag{C3}$$

On any finite interval, the supremum norm of $e^{kx}X(x)$ in (C2) is bounded by $1/(1-\eta_f)$. Substituting this bound into the convolution gives

$$0 \leq X(x) - F(x) \leq \frac{\eta_f}{1-\eta_f} e^{-kx} \leq 100\chi e^{-2x/3}. \tag{C4}$$

Consequently, for sufficiently small χ,

$$\begin{aligned}(-q)_+ &\leq \sup_{x\geq 0}[100\chi e^{-2x/3} - e^{-x}]_+ \\ &\leq \frac{4}{27}(100\chi)^3 = O(\nu^{3/2}).\end{aligned} \tag{C5}$$

This weighted-kernel argument is uniform at coincident decay rates and as $\tau \to \infty$.

For $0 < \tau \leq 1/2$, define

$$\begin{aligned}d_{\mathrm{sp}} &= \sqrt{(1-\zeta)^2 + 4h_f}, \\ \beta &= \frac{1+\zeta-d_{\mathrm{sp}}}{2}, \qquad \beta_+ = \frac{1+\zeta+d_{\mathrm{sp}}}{2}, \\ d_0 &= \frac{d_{\mathrm{sp}}+\zeta-1}{2d_{\mathrm{sp}}}.\end{aligned} \tag{C6}$$

For small ν, $0 < \beta < 1 \leq \beta_+$ and $0 < d_0 < 1$. The exact comparison solution is

$$X(x) = (1-d_0)e^{-\beta_+ x} + d_0 e^{-\beta x}.$$

Equation (C2) therefore gives $q \geq (1+d_0)e^{-x} - d_0 e^{-\beta x}$. Maximizing its negative part over $x \geq 0$ gives

$$(-q)_+ \leq \mathcal{D}_0(\beta, d_0) := (1-\beta)d_0 \left(\frac{d_0\beta}{1+d_0}\right)^{\beta/(1-\beta)}. \tag{C7}$$

Because $1-\zeta \geq 1/4$ throughout this ratio range, the expansions

$$\beta = \frac{1+\tau}{2} + O(\chi), \qquad d_0 = \frac{8\chi\tau}{(1-\tau)^2}[1+O(\chi)] \tag{C8}$$

are uniform. They follow, for example, by rationalizing $d_{\mathrm{sp}} - (1-\zeta) = 4h_f/[d_{\mathrm{sp}} + (1-\zeta)]$.

2. Uniform asymptotics

Here C_{d} denotes a finite universal constant, independent of the rates and ν.

If $0 < \tau \leq \sqrt{\chi}$, then $d_0 \leq C_{\mathrm{d}}\chi^{3/2}$ and $\beta \geq 1/3$ for small χ. Hence

$$\begin{aligned}\mathcal{D}_0(\beta, d_0) &\leq d_0^{1/(1-\beta)} \leq d_0^{3/2} = O(\chi^{9/4}) \\ &= o\big(\nu/\log^2(1/\nu)\big).\end{aligned} \tag{C9}$$

This bound controls the smallest rate ratios before the expansion involving $\log\tau$.

For the remaining range $\sqrt{\chi} \leq \tau \leq 1/2$, put $L = \log(1/\nu)$. Then $|\log\tau| = O(L)$ and $\chi L \to 0$. The sensitivity identity

$$\partial_\beta \log \mathcal{D}_0(\beta, d_0) = \frac{\log[d_0\beta/(1+d_0)]}{(1-\beta)^2} \tag{C10}$$

shows that the $O(\chi)$ correction to β in (C8) changes $\log\mathcal{D}_0$ by a uniformly $o(1)$ amount. The relative $O(\chi)$ correction to d_0 also has an $o(1)$ effect, since $\partial_{\log d_0}\log\mathcal{D}_0$ is bounded in this range. Using $\chi^2 = \nu/(1-\nu)$ gives

$$\frac{\mathcal{D}_0(\beta, d_0)}{\nu} = [1+o(1)]\mathcal{F}(\tau)\exp\left[-\frac{\tau L}{1-\tau}\right], \tag{C11}$$

where

$$\mathcal{F}(\tau) = \frac{1-\tau}{2}\left(\frac{1+\tau}{2}\right)^{(1+\tau)/(1-\tau)}\left(\frac{8\tau}{(1-\tau)^2}\right)^{2/(1-\tau)}. \tag{C12}$$

The function has a bounded continuous extension to $[0, 1/2]$, with

$$\mathcal{F}(\tau) = 16\tau^2\{1 + O[\tau(1+|\log\tau|)]\}. \tag{C13}$$

For $\tau \geq L^{-1/2}$, (C11) is exponentially small in $\sqrt{L}$. On the remaining range, the relative error in (C13) tends uniformly to zero. As $1/(1-\tau) \geq 1$,

$$\begin{aligned}\sup_\tau \frac{\mathcal{D}_0(\beta, d_0)}{\nu} &\\ &\leq [1+o(1)]\sup_{\tau>0} 16\tau^2 e^{-\tau L} + o(L^{-2}) \\ &= \frac{64/e^2 + o(1)}{L^2}.\end{aligned} \tag{C14}$$

Together with (C5) and (C9), this proves a uniform bound for all finite generators in the restricted class. For $q < 0$, Eq. (29) gives $\nu \leq \epsilon$. Since $x/\log^2(1/x)$ increases for small positive x, the same leading bound holds with ϵ in place of ν; for $q \geq 0$, it is immediate. The resulting bound depends only on the measured probabilities and extends to $\mathcal{Q}_{\mathrm{diag}}$ as in Section V.

3. Matching generator

Take $\nu \downarrow 0$, $\chi = \sqrt{\nu/(1-\nu)}$, $L = \log(1/\nu)$, and $\tau = 2/L$. Let $B = \tau A$, $U = \chi^2 A$, and in the measurement

basis choose

$$J = \begin{pmatrix} -\sqrt{B} & -\sqrt{U} \\ \sqrt{A} & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} 0 & i\kappa_H \\ -i\kappa_H & 0 \end{pmatrix}, \qquad \kappa_H = \frac{\sqrt{AB}(\nu + \chi/2)}{1 - 2\nu}. \tag{C15}$$

Equation (A6) confirms that $\mathrm{diag}(\nu, 1-\nu)$ is stationary. The coefficients satisfy $\widetilde{C} = -c_0$, $R_0 = r_0$, $E = -\sqrt{AU}$, and $\omega = 0$. The subspace with real coherence is invariant and obeys (31). This single-jump generator realizes the negative-feedback mechanism described in Section IV.

Write $\zeta_+ = \lambda_+/\alpha$ and define

$$\Delta_- = \sqrt{(1-\zeta_+)^2 - 4h_f}, \qquad \gamma_\pm = \frac{1 + \zeta_+ \pm \Delta_-}{2}, \qquad d_- = \frac{1 - \zeta_+ - \Delta_-}{2\Delta_-}. \tag{C16}$$

For $\tau = 2/L$ and small ν, these quantities are real, with $0 < \gamma_- < \gamma_+$ and $d_- > 0$. The exact response is

$$q(x) = (1 + d_-)e^{-\gamma_+ x} - d_- e^{-\gamma_- x}, \qquad x = \alpha. \tag{C17}$$

Choose the finite positive scale

$$x_* = \frac{\log[(1+d_-)\gamma_+/(d_-\gamma_-)]}{\gamma_+ - \gamma_-}, \qquad A = \frac{x_*}{1+\chi^2}. \tag{C18}$$

Then the achieved negative determinant is $-q(x_*) = \mathcal{D}_0(\beta_-, d_-)$ with $\beta_- = \gamma_-/\gamma_+$. These parameters have the same leading expansions as (C8). Substitution of $\tau = 2/L$ into (C11)–(C13) therefore gives

$$-q(x_*) = \left(\frac{64}{e^2} + o(1)\right)\frac{\nu}{L^2}. \tag{C19}$$

The actual disturbance is $\epsilon(\nu) = \nu[1 - q(x_*)] \sim \nu$. It is positive, continuous, and tends to zero, so the connected-range argument used for (27) covers every sufficiently small positive ϵ. This matches the upper bound (C14) and completes Theorem 2. Figure 3 shows both constructions approaching their asymptotic constants.

The curves in Figs. 1 and 3 evaluate the explicit solutions with 100-digit arithmetic. Figure 1 matches the two constructions at common actual errors by solving for their family parameters; Fig. 3 uses the actual error of each sampled realization for both axes.

4. Prescribed stationary-coherence fraction

The physical constraints in Eqs. (17) and (19) can be attained at each fixed fraction of the leading reversal. Fix $\eta \in (0, 1]$, let $L \to \infty$, and set

$$\sigma = e^{-L}, \qquad p = \eta\sigma, \qquad \nu = (1-\eta)\sigma, \qquad \tau = L^{-1}, \qquad A = L + \log L + 1 - \log(4\eta), \qquad B = \tau A. \tag{C20}$$

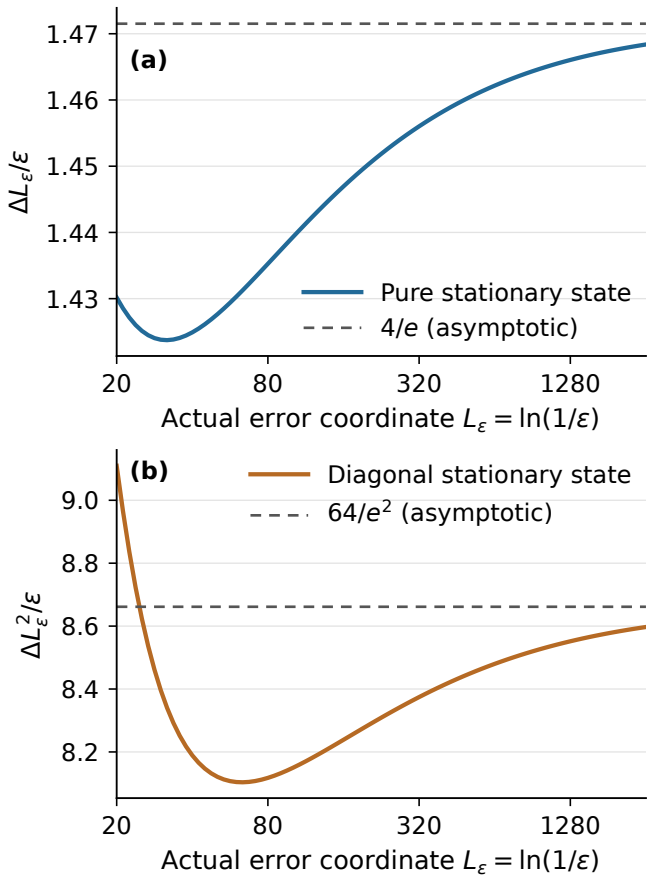


FIG. 3. Convergence of the explicit reset constructions, with population-order reversal $\Delta = \epsilon - a$ and actual disturbance $\epsilon = P(0|1)$. (a) The pure-stationary-state family, Eqs. (21)–(23), approaches $4/e$ after logarithmic rescaling. (b) The diagonal-stationary-state family, Eqs. (C15)–(C18), with $p = 0$, approaches $64/e^2$ after rescaling by $\log^2(1/\epsilon)$. Solid curves evaluate the explicit solutions; dashed lines show the asymptotic constants. The horizontal axes are logarithmic.

Use the basis in Eq. (22) and the matrices in Eq. (C15), now expressed in that stationary eigenbasis rather than the measurement basis. This defines a finite GKSL generator with stationary state $\nu|e\rangle\langle e| + (1-\nu)|g\rangle\langle g|$. Writing $\chi = \sqrt{\nu/(1-\nu)}$, the real deviations obey $d(u,z)^T/dt = AM_0(u,z)^T$, where

$$M_0 = \begin{pmatrix} -\alpha_0 & b_0 \\ c_0 & -\beta_0 \end{pmatrix}, \qquad \alpha_0 = 1 + \chi^2, \quad \beta_0 = \frac{(1+\chi)^2 + \tau}{2}, \qquad b_0 = \frac{2\sqrt{\tau}\chi}{1-\chi}, \quad c_0 = -\frac{\sqrt{\tau}(1+\chi^2)}{1-\chi}. \tag{C21}$$

The imaginary coherence decays at rate $A[(1-\chi)^2 + \tau]/2$. Both real decay rates are positive for sufficiently large L, so the stationary state is unique. Let $g_0 = \sqrt{(\alpha_0 - \beta_0)^2 + 4b_0c_0}$ and $\lambda_{f,s} = (\alpha_0 + \beta_0 \pm g_0)/2$. With

v, w from Eq. (20), the exact response is

$$q = (1-Q_s)e^{-\lambda_f A} + Q_s e^{-\lambda_s A},$$
$$Q_s = \frac{(\lambda_f-\alpha_0)v^2 + (b_0+c_0)vw + (\lambda_f-\beta_0)w^2}{g_0}. \quad \text{(C22)}$$

For each fixed $\eta > 0$, $\lambda_f = 1+O(\chi\tau+\chi^2)$, $\lambda_s = (1+\tau)/2+O(\chi)$, and $Q_s = -4\sqrt{p\tau}[1+O_\eta(L^{-1/2})]$. Substituting Eq. (C20) gives

$$q = -\left(\frac{4\eta}{e} + o(1)\right)\frac{\sigma}{L}. \quad \text{(C23)}$$

In particular, the mixed-state feedback contributes to the remainder without changing this leading coefficient.

For $K = e^{AM_0}$, the actual target-input disturbance is

$$\epsilon(L) = \nu + (1-2\nu)p + (v,w)K\begin{pmatrix} p-\nu \\ -w/2\end{pmatrix}. \quad \text{(C24)}$$

The spectral projectors give $K_{00}, K_{01} = O_\eta(\sigma/L)$, $K_{10} = O_\eta(\sqrt{\sigma}/L)$, and $K_{11} = O_\eta(\sqrt{\sigma/L})$. Therefore $\epsilon(L) \sim \sigma$ and $\log(1/\epsilon(L)) \sim L$, proving

$$r_{\epsilon(L)} \to \eta, \qquad \frac{C_*}{\epsilon(L)} \to \eta, \qquad \frac{I_*}{2\epsilon(L)} \to 1-\eta. \quad \text{(C25)}$$

The same matrix estimates, together with the imaginary-coherence decay, imply $\sup_\rho \frac{1}{2}\|\Phi(\rho)-\rho_*\|_1 = O_\eta(\sqrt{\sigma/L})$. Since $\frac{1}{2}\|\rho_* - |1\rangle\langle 1|\|_1 \sim \sqrt{\eta\sigma}$, the stationary input and triangle inequality yield $E_{\mathrm{tr}}^2/\epsilon \to \eta$.

The function $\epsilon(L)$ is continuous, positive on a sufficiently large closed tail, and tends to zero. The connected-range argument following Eq. (27) therefore supplies a finite realization at every sufficiently small actual ϵ, with the limits above preserved. For $\eta = 0$, the diagonal construction gives $C_* = 0$, $I_*/(2\epsilon) \to 1$, and $r_\epsilon \to 0$. These statements concern fixed η; no uniform crossover as η tends to zero with ϵ is asserted.

Appendix D: Relation to the four-parameter family

This appendix establishes the relation stated in Section VI between the attaining constructions and the family used for numerical boundary searches in Ref. [1]. That family has

$$H_{\mathrm{old}} = X, \qquad J_{\mathrm{old}} = \sqrt{\gamma}\,|\psi_o\rangle\langle\psi_i|,$$
$$|\psi_j\rangle = \begin{pmatrix}\cos\theta_j \\ i\sin\theta_j\end{pmatrix}, \qquad \gamma, t > 0. \quad \text{(D1)}$$

The rate and duration are unrestricted parameters of this family.

For each fixed finite parameter choice of the pure-stationary-state construction,

$$J = \sqrt{A+B}\,|u\rangle\langle e|, \qquad |u\rangle = \frac{-\sqrt{B}|e\rangle + \sqrt{A}|g\rangle}{\sqrt{A+B}}. \quad \text{(D2)}$$

The vectors u, e are real in the measurement basis. The unitary $S = \mathrm{diag}(1, i)$ preserves the preparation and measurement projectors and maps any normalized real vector to the form in (D1). Set $V_{\mathrm{old}} = S|u\rangle\langle e|S^\dagger$. For $t_n \downarrow 0$ and $\gamma_n t_n = A+B$, the channels in the prior family satisfy

$$\exp(-it_n[X,\cdot] + (A+B)\mathcal{D}[V_{\mathrm{old}}]) \longrightarrow \exp\big(\mathcal{D}[SJS^\dagger]\big). \quad \text{(D3)}$$

Continuity of the finite-dimensional matrix exponential places each such probability table in the closure of the prior family. This approximation holds at fixed A, B, p; the subsequent limit $L \to \infty$ approaches reset.

For the diagonal construction (C15), $\tau = 2/\log(1/\nu) > 4\chi$ for sufficiently small ν. The real jump then has a negative real eigenvalue ξ satisfying

$$\xi^2 + \sqrt{B}\,\xi + A\chi = 0, \qquad J' = J - \xi I,$$
$$H' = H + \frac{i\xi}{2}(J - J^\dagger). \quad \text{(D4)}$$

This identity shift preserves the GKSL generator. Since J' is nonzero, real, and singular, it has rank one. Moreover,

$$H'_{12} = ik',$$
$$k' = \kappa_H - \frac{\xi}{2}(\sqrt{A}+\sqrt{U}) > 0, \quad \text{(D5)}$$
$$SH'S^\dagger = k'X.$$

Write $J' = s|u'\rangle\langle v'|$ with $s > 0$ and real unit vectors. After the same phase change, choose duration $t = k'$ and jump rate $\gamma = s^2/k'$ in the prior family. The integrated generator then equals the transformed time-one generator of the diagonal construction. Thus the diagonal family belongs to the prior ansatz for every sufficiently small ν.

Appendix E: Comparison of exclusion regions

To compare with the earlier analytic outer bound, denote the reset-side function in Theorem 1 of Ref. [1] by

$$g_R(a) = (2-a)(2a + a^{1/10}). \quad \text{(E1)}$$

That theorem excludes tables when $a \le 10^{-6}$, $\epsilon > g_R(a)$, and a second condition separating them from the swap endpoint holds. Near reset that second condition holds automatically. Since $g_R(a) = 2a^{1/10}[1+o(1)]$ is locally increasing, its inversion gives the necessary lower bound

$$a \ge (\epsilon/2)^{10}[1+o(1)]. \quad \text{(E2)}$$

Equation (E2) is the necessary condition obtained by inverting the earlier exclusion bound.

To compare the excluded regions directly, fix $C > 4/e$ and choose $\epsilon_C > 0$ so that Theorem 1 gives $a \ge \epsilon[1 - C/\log(1/\epsilon)]$ below this threshold. Every table in

$$\mathcal{W}_C = \big\{(a, 1-\epsilon) : 0 < \epsilon < \min(\epsilon_C, e^{-4C}, 10^{-6}),$$
$$\epsilon/2 < a < 3\epsilon/4\big\} \quad \text{(E3)}$$

is excluded by the present bound. None is excluded by the earlier theorem or its exchanged-coordinate version. Indeed, for $0 < a < 1$, $g_R(a) > 2a(2-a) > 2a > \epsilon$, so the earlier reset condition fails. After exchanging the coordinates, the first coordinate is near one and fails the small-coordinate premise. The additional excluded region therefore has nonzero area.

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